

KATHMANDU MODEL SECONDARY SCHOOL

Second Term Examination 2080

SET - B

Subject: Physics '1011'

Time: 20 mins.

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions:

GROUP 'A' [11×1=11]

Choose the best alternative to the following questions.

Write the dimensions of $\frac{b}{a}$ in the relation, $P = \frac{b-x^2}{at}$ where, P is power, x is the distance and t the time.

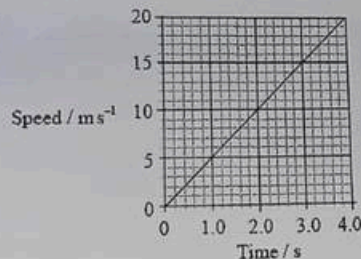
- a) $[M^{-1} L^0 T^2]$
b) $[M^{-1} L^{-2} T^2]$

- c) $[M^0 L^2 T^0]$
d) $[M^1 L^2 T^2]$

The graph is a speed versus time graph for an object that is moving in a straight line.

The distance travelled by the object during the first 4.0 seconds is

- a) 80 m. b) 40 m.
c) 20 m. d) 5 m.



A car drives 10 miles towards the northeast direction and then 6 miles towards the southeast direction. What is the magnitude of the displacement of the car from its starting point?

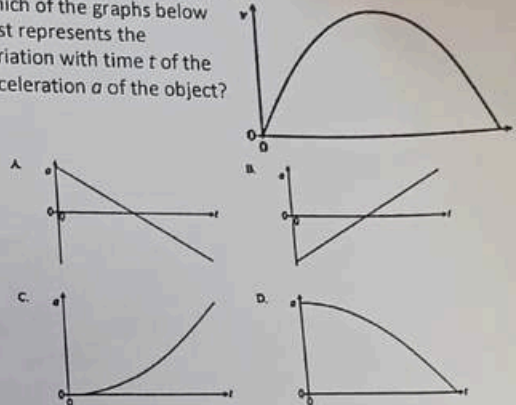
- a) 16 miles b) 4 miles c) 14 miles d) $\sqrt{136}$ miles
A particle has an initial velocity of $3\hat{i} + 4\hat{j}$ and an acceleration of $4\hat{i} + 0.3\hat{j}$. Its speed after 10 s is
a) 10 units b) 7 units c) $7\sqrt{2}$ units d) 8.5 units

A and B are two hollow, concentric, conducting spheres, with A inside B. A and B are given charges Q_1 and Q_2 respectively. If they are now joined by a thin wire, what charge will flow from B to A?

- a) $Q_1 - Q_2$ b) $(Q_1 - Q_2)/2$ c) $(Q_1 + Q_2)/2$ d) 0

10. In the electrolysis of fused salt, the amount of the substance deposited at the cathode will not depend on
a) current intensity b) time of electrolysis
c) temperature of the both d) electrochemical equivalent
11. The oxide which cannot act as a reducing agent is
a) NO_2 b) CO_2 c) SO_2 d) ClO_2

6. The graph below shows the variation with time t of the velocity v of an object moving on a straight-line. Which of the graphs below best represents the variation with time t of the acceleration a of the object?



7. A ray of light travelling in a transparent medium falls on a surface separating the medium from air, at an angle of incidence of 40° . The ray undergoes total internal reflection. Select the possible values of refractive index of the medium with respect to air, from the following.
a) 1.3 b) 1.4 c) 1.5 d) 1.6
8. If the object is real, a convex mirror always forms
a) Real and inverted image b) real and erect image
c) virtual and erect image d) virtual and inverted image
9. The freezing point on a thermometer is marked as -20° and the boiling point as 130° . A temperature of human body ($34^\circ C$) on this thermometer will be read as
a) 31° b) 51° c) 20° d) None of these
10. Two copper rods P and Q of lengths 5m and 8m respectively are heated from $30^\circ C$ to $100^\circ C$. The ratio of increase in length of rod P to Q is
a) $\frac{3}{4}$ b) $\frac{1}{2}$ c) 1 d) $\frac{5}{8}$
11. 200g of water at $48^\circ C$ is mixed with 100g of water at $30^\circ C$. The resulting temperature of water is
a) $45^\circ C$ b) $42^\circ C$ c) $39^\circ C$ d) $36^\circ C$

KATHMANDU MODEL SECONDARY SCHOOL

Second Term Examination 2080

SET - A

Subject: Physics '1011'

Time: 20 mins.

Candidates are required to give their answers in their own words as far as possible. The figures in the margin indicate full marks.

Attempt all the questions:

GROUP 'A' [11×1=11]

Choose the best alternative to the following questions.

Write the dimensions of $\frac{a}{b}$ in the relation, $P = \frac{b-x^2}{at}$ where, P is power, x the distance and t the time.

a) $[M^{-1} L^0 T^2]$

b) $[M^0 L^2 T^0]$

c) $[M^{-1} L^{-2} T^2]$

d) $[M^1 L^2 T^{-2}]$

The graph shows the variation with time t of the acceleration a of an object.

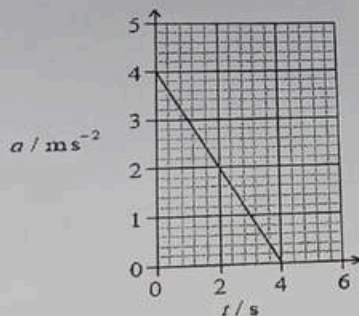
Which of the following is the change in velocity of the object in the time interval 0 to 4 s?

a) -8 m s^{-1}

b) -4 m s^{-1}

c) $+4 \text{ m s}^{-1}$

d) $+8 \text{ m s}^{-1}$



A man walks 8 m towards east and 6 m towards north. The magnitude of displacement is

a) 14 m

b) 2 m

c) 10 m

d) $(8^2 + 6^2)$

An object starts with an initial velocity of $5\hat{i} + 12\hat{j}$ and experiences an acceleration of $-1.5\hat{i} - 2\hat{j}$. What is the speed of the object after 8 seconds?

a) 10 units

b) 15 units

c) 13 units

d) 17 units

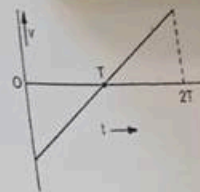
5. The figure shows the velocity (v) of a particle plotted against time (t).

a) The particle changes its direction of motion at some point.

b) The acceleration of the particle remains constant.

c) The displacement of the particle is zero.

d) The initial and final speeds of the particle are the same.



6. A and B are two hollow, concentric, conducting spheres, with A inside B. A and B are given charges Q_1 and Q_2 respectively. If they are now joined by a thin wire, what charge will flow from A to B?

a) Q_1

b) $(Q_1 - Q_2)/2$

c) $Q_2/2$

d) $Q_1/2$

7. A ray of light travelling in a transparent medium falls on a surface separating the medium from air, at an angle of incidence of 45° . The ray undergoes total internal reflection. Select the possible values of refractive index of the medium with respect to air, from the following.

a) 1.3

b) 1.4

c) 1.5

d) 1.2

8. An object is placed at a distance twice the focal length of concave mirror. Then image formed is

a) virtual and inverted

b) real and erect

c) real, inverted and diminished

d) real, inverted and of same size as object

9. The absolute zero temperature in fahrenheit scale is

a) -273°F

b) -32°F

c) -460°F

d) -132°F

10. Two copper rods P and Q of lengths 1 m and 4 m respectively are heated from 20°C to 100°C . The ratio of increase in length of rod P to Q is

a) 2

b) $\frac{1}{2}$

c) 4

d) $\frac{1}{4}$

11. 420 J of heat is used in heating 10g of water. The temperature of water then increases by (specific heat capacity of water = $4.2 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$)

a) 10°C

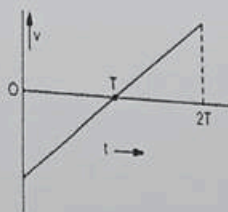
b) 42°C

c) 4.2°C

d) 100°C

The figure shows the velocity (v) of a particle plotted against time (t).

- The particle changes its direction of motion at some point.
- The acceleration of the particle remains constant.
- The displacement of the particle is zero.
- The initial and final speeds of the particle are the same.



A and B are two hollows, concentric, conducting spheres, with A inside B. A and B are given charges Q_1 and Q_2 respectively. If they are now joined by a thin wire, what charge will flow from A to B?

- Q_1
- $(Q_1 - Q_2)/2$
- $Q_2/2$
- $Q_1/2$

A ray of light travelling in a transparent medium falls on a surface separating the medium from air, at an angle of incidence of 45° . The ray undergoes total internal reflection. Select the possible values of refractive index of the medium with respect to air, from the following.

- 1.3
- 1.4
- 1.5
- 1.2

An object is placed at a distance twice the the focal length of concave mirror. Then image formed is

- virtual and inverted
- real and erect
- real, inverted and diminished
- real, inverted and of same size as object

The absolute zero temperature in fahrenheit scale is

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- $\frac{1}{2}$
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- $\frac{1}{4}$

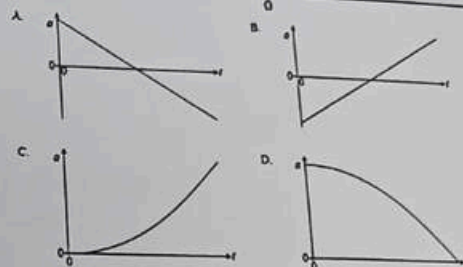
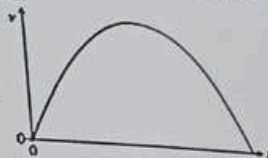
420 J of heat is used in heating 10g of water. The temperature of water then increases by (specific heat capacity of water = $4.2 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$)

- 10°C
- 42°C
- 4.2°C
- 100°C

- electrode will not depend on
- current intensity
 - time of electrolysis
 - temperature of the both
 - electrochemical equivalent of ions
11. The oxide which cannot act as a reducing agent is
- NO_2
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 - SO_2
 - ClO_2

6. The graph below shows the variation with time t of the velocity v of an object moving on a straight-line.

Which of the graphs below best represents the variation with time t of the acceleration a of the object?



7. A ray of light travelling in a transparent medium falls on a surface separating the medium from air, at an angle of incidence of 40° . The ray undergoes total internal reflection. Select the possible values of refractive index of the medium with respect to air, from the following.

- 1.3
- 1.4
- 1.5
- 1.6

8. If the object is real, a convex mirror always forms

- Real and inverted image
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- virtual and inverted image

9. The freezing point on a thermometer is marked as -20° and the boiling point as 130° . A temperature of human body (34°C) on this thermometer will be read as

- 31°
- 51°
- 20°
- None of these

10. Two copper rods P and Q of lengths 5m and 8m respectively are heated from 30°C to 100°C . The ratio of increase in length of rod P to Q is

- $\frac{3}{4}$
- $\frac{1}{2}$
- 1
- $\frac{5}{8}$

11. 200g of water at 48°C is mixed with 100g of water at 30°C . The resulting temperature of water is

- 45°C
- 42°C
- 39°C
- 36°C

KATHMANDU MODEL SECONDARY SCHOOL

Second Term Examination 2080

SET - A

Subject: Physics '1011'

Class: XI 'Science'

Time: 20 mins.

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions:

GROUP 'A' [11×1=11]

Choose the best alternative to the following questions.

Write the dimensions of $\frac{a}{b}$ in the relation, $P = \frac{b-x^2}{at}$ where, P is power, x the distance and t the time.

a) $[M^{-1} L^0 T^2]$

b) $[M^0 L^2 T^0]$

c) $[M^{-1} L^{-2} T^2]$

d) $[M^1 L^2 T^{-2}]$

The graph shows the variation with time t of the acceleration a of an object.

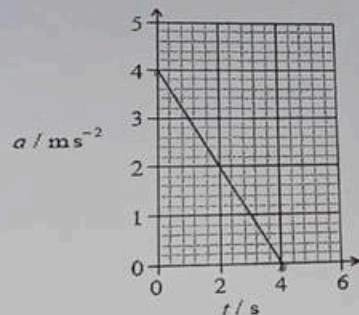
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A man walks 8 m towards east and 6 m towards north. The magnitude of displacement is

a) 14 m

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An object starts with an initial velocity of $5\hat{i} + 12\hat{j}$ and experiences an acceleration of $-1.5\hat{i} - 2\hat{j}$. What is the speed of the object after 8 seconds?

a) 10 units

b) 15 units

c) 13 units

d) 17 units

KATHMANDU MODEL SECONDARY SCHOOL

Second Term Examination 2080

SET - B

Class: XI 'Science'

Time: 20 mins.

Subject: Physics '1011'

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions:

GROUP 'A' [11×1=11]

Choose the best alternative to the following questions.

1. Write the dimensions of $\frac{b}{a}$ in the relation, $P = \frac{b-x^2}{at}$ where, P is power, x the distance and t the time.

a) $[M^{-1} L^0 T^2]$

b) $[M^0 L^2 T^0]$

c) $[M^{-1} L^{-2} T^2]$

d) $[M^1 L^2 T^{-2}]$

2. The graph is a speed versus time graph for an object that is moving in a straight line.

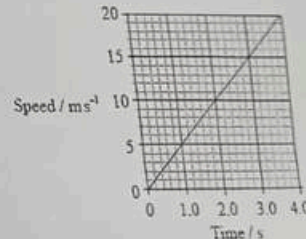
The distance travelled by the object during the first 4.0 seconds is

a) 80 m.

b) 40 m.

c) 20 m.

d) 5 m.



3. A car drives 10 miles towards the northeast direction and then 6 miles towards the southeast direction. What is the magnitude of the displacement of the car from its starting point?

a) 16 miles

b) 4 miles

c) 14 miles

d) $\sqrt{136}$ miles

4. A particle has an initial velocity of $3\hat{i} + 4\hat{j}$ and an acceleration of $0.4\hat{i} + 0.3\hat{j}$. Its speed after 10 s is

a) 10 units

b) 7 units

c) $7\sqrt{2}$ units

d) 8.5 units

5. A and B are two hollow, concentric, conducting spheres, with A inside B. A and B are given charges Q_1 and Q_2 respectively. If they are now joined by a thin wire, what charge will flow from B to A?

a) $Q_1 - Q_2$

b) $(Q_1 - Q_2)/2$

c) $(Q_1 + Q_2)/2$

d) 0

- b) A ball is dropped from a height 20 m and rebounds with a velocity $\frac{1}{2}$ of the velocity with which it hits the ground. What is the time interval between first and second bounces?

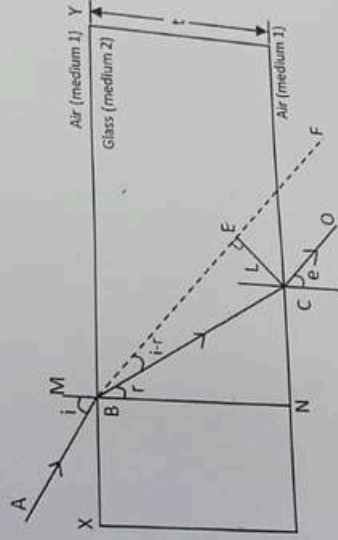
GROUP 'C' [3×8=24]

1. a) A body is projected from ground making angle θ with horizontal. Show that it's path is parabolic. Also derive expressions for time of flight and maximum height. [3]
- b) Can two different angles of projections achieve same horizontal range? Justify your answer. [2]
- c) After winning the Cricket World Cup, Pat Cummins, the Australian captain, pops a corked fruit wine. If the cork shoots out of the bottle at an angle of 40° to the horizontal and travels a horizontal distance of 4.50 meters in 1.25 seconds, calculate the initial speed of the cork. [3]

- ... achieves same horizontal range? Justify your answer.

- the cork shoots out of the bottle at an angle of 40° to the horizontal and travels a horizontal distance of 4.50 meters in 1.25 seconds, calculate the initial speed of the cork.

2. A ray of light AB falls in a glass slab at an angle of incidence 'i' and is refracted along BC with 'r' as angle of refraction and finally emerges along CO with 'e' as angle of emergence, 't' is the thickness of glass slab. Answer the following questions.



- b)) A microscope is focused on the upper surface of a glass plate. A second plate is then placed over the first. To focus on the bottom surface of the second plate, the microscope must be raised 1mm. To focus on the upper surface, it must be raised 2mm farther. Find the index of refraction of the second plate. [2]

-fection of the second plate.

- c) An underwater swimmer is at a depth of 12 meters below the water surface. A bird is positioned 18 meters above the water surface, directly in the line of sight of the swimmer's eyes. Find the apparent distance of the bird from the water surface as seen by the swimmer. [2]

- line with the swimmer. (Refractive Index of water is $\frac{4}{3}$), and the electric field intensity due to a long [1+3]

3. a) State Gauss theorem in electrostatics and use it to find the electric field of a uniformly charged infinite plane. [3]

- a) **straight charged conductor.** $\vec{E} = 3 \times 10^3 \hat{i} \text{ N/C}$. What is the electric field distributed uniformly on its plane?

- b) Consider a uniform electric field of magnitude $1.6 \times 10^6 \text{ N/C}$ directed parallel to the x -axis. A rectangular conductor of radius 12 cm has a charge of $1.6 \times 10^{-6} \text{ C}$ distributed uniformly on its surface. Find the electric field at the center of the sphere?

- of 10 cm on a side. (a) What is the electric field? (b) What is the electric flux through the surface of the sphere?

- c) A spherical conductor of radius R carries a positive charge Q . What is the electric field E at a point (i) inside the sphere, (ii) at a point outside the sphere?

Class: XI 'Science'

Time: 3hrs.

Subject: Physics '1011'

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

F.M.: 75

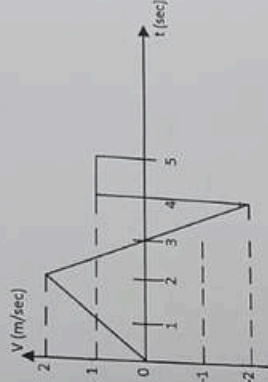
P.M.: 30

Attempt all the questions:

GROUP 'B' [8×5=40]

Short answer questions

1. a) The velocity time graph is shown in the fig. below calculate the displacement and distance from the graph at time 5 sec. [2]
- b) If a body starts from rest and travels 120 m in the 8th minute, then what is its acceleration? [2]
- c) Can any two rectangular components of a given vector have a magnitude greater than vector itself? [1]
2. a) Derive an equation relating magnification and focal length of a curved mirror. [2]
- b) A child looking in to a small hemispherical metal bowl sees an erect image of himself 21 cm from the bowl. He turns the bowl over and sees a further erect image of himself 7 cm from the bowl. What is the radius of curvature of the bowl? [3]
3. a) Why is calorimeter made up of copper and lagged with a wooden box? [2]
- b) A tap water supplies water at 10°C and another tap at 100°C. How much hot water must be taken so that we get 20 kg of water at 35°C? [3]
4. a) Explain the variation of density of substance with temperature. [2]
- b) Describe with the mathematical details to determine the real expansivity of liquid. [3]
5. a) Explain the concept of water equivalent of a substance. [2]
- b) What is the meaning of heat capacity of 0.5 Kg of copper = 195 J K^{-1} ? From this information, find the final temperature of mixture when, in copper vessel of mass 100 gm at 100°C, 50 gm of water at 25 °C is poured. (specific heat capacity of water $4.2 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$) [3]
6. a) Define electric potential. Derive an expression for electric potential at a point in the electrostatic field of a point charge. [2]
- b) An electron is situated in uniform electric field of intensity $1.2 \times 10^5 \text{ N/C}$. Find the time it takes to travel 4 cm starting from rest. [1]
7. a) A metal rod held in hand and rubbed with wool does not show any sign of being charged. Explain why? [1+1]
- b) Define electric lines of force. Why two electric lines of force never intersect? [1+1]
- c) Four-point charges $q_A = 2\mu\text{C}$, $q_B = -5\mu\text{C}$, $q_C = 2\mu\text{C}$ and $q_D = -5\mu\text{C}$ are located at the corners of a square ABCD of side 10 cm in order. What is the force on a charge of $1\mu\text{C}$ placed at centre of the square?



- 13) a) State and use parallelogram law of vector addition to find the magnitude and direction of the resultant of two vectors inclined at an angle θ . [3]
 b) Two forces F_1 and F_2 act at a point. F_1 has magnitude 8N and acts at an angle of 30° with the positive x-axis. F_2 has a magnitude of 6N and acts at an angle of 150° with positive x-axis. Find the magnitude and direction of resultant force. [3]
- 14) a) Derive a relation between focal length and radius of curvature of a spherical mirror. [2]
 b) A small candle 2.5 cm in size is placed 27 cm in front of a concave mirror of radius of curvature 36 cm. At what distance from the mirror should a screen be placed in order to receive a sharp image? Describe the nature & size of the image. [2]
- 15) a) Derive a mirror formula for concave mirror and mention the sign convention. [3]
 b) A concave mirror forms an image 3 times as large as an object placed 0.3 m from it. Calculate its focal length. [3]
- 16) a) Define the coefficient linear, superficial and cubical expansions. Derive relations between them. [4]
 b) Why a thick glass cracks when boiling water is poured in it? [1]
- 17) a) Pendulum clocks generally go fast in winter and slow in summer. Why? [1]
 b) A pendulum clock loses 12 Sec. a day if the temperature is 40°C and gains 4 Sec. a day if the temperature is 20°C . Find the temperature at which the clock will show correct time and coefficient of linear expansion of the metal of pendulum shaft. [4]
- 18) a) State and explain the principle of superposition of Coulomb's force. [2]
 b) Two-point charges of $10\ \mu\text{C}$ and $-40\ \mu\text{C}$ are separated by distance of 13 cm. Find the point along the line joining them at which net electrostatic force on any charge is zero. [3]
- 19) a) Define dielectric constant. Why its value is always greater than one? [2]
 b) Four equal charges q are placed at the four corners of a square of length a . Find the magnitude of the net force acting on one of them due to other charges and draw diagram to show the direction of force on that charge. [3]

GROUP 'C' [3×8=24]

Long answer questions:

- 20) a) Explain why vector addition is not commutative, providing a real-life example to support your reasoning. [2]
 b) How will you resolve a vector in a plane in two mutually perpendicular directions? [3]
 c) A pilot flies 200 km at 60° north of east, then 150 km at 30° south of west and finally 100 km at 45° south of east. Calculate the magnitude and direction of the resultant displacement. [1]
- 21) a) Why does a solid expand on heating? [3]
 b) Explain an experiment to determine the coefficient of linear expansion of a solid. [1]
 c) A steel cylinder of a car engine has an aluminium piston. The internal diameter of the cylinder at 15°C is 8.0 cm and there is an all-round clearance of 0.05 mm between the piston and the cylinder of the wall. Find the temperature at which the two will exactly fit in. [4]
 ($\alpha_{\text{steel}} = 1.2 \times 10^{-5}\ \text{C}^{-1}$ and $\alpha_{\text{aluminium}} = 1.6 \times 10^{-5}\ \text{C}^{-1}$) [3]
- 22) a) State and explain Coulomb's law in electrostatics. [3]
 b) Why Coulomb's law is applicable to stationary charges only? [1]
 c) Define 1 coulomb charge on the basis of Coulomb's law. [1]
 d) Two charges $10^{-6}\ \text{C}$ and $10^{-7}\ \text{C}$ attract each other with a force of 400N. Calculate the distance between them if they are in [3]
 i) air and
 ii) water (relative permittivity of water is 80)

- 13) a) State and use parallelogram law of vector addition to find the magnitude and direction of the resultant of two vectors inclined at an angle θ . [3]
 b) Two forces F_1 and F_2 act at a point. F_1 has magnitude 8N and acts at an angle of 30° with the positive x-axis. F_2 has a magnitude of 6N and acts at an angle of 150° with positive x-axis. Find the magnitude and direction of resultant force. [3]
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 i) A steel cylinder of a car engine has an aluminium piston. The internal diameter of the cylinder of the wall. [1]
 j) A steel cylinder of a car engine has an aluminium piston. The internal diameter of the cylinder of the wall. [1]
 k) A steel cylinder of a car engine has an aluminium piston. The internal diameter of the cylinder of the wall. [1]
 l) A steel cylinder of a car engine has an aluminium piston. The internal diameter of the cylinder of the wall. [1]
 m) A steel cylinder of a car engine has an aluminium piston. The internal diameter of the cylinder of the wall. [1]
 n) A steel cylinder of a car engine has an aluminium piston. The internal diameter of the cylinder of the wall. [1]
 o) A steel cylinder of a car engine has an aluminium piston. The internal diameter of the cylinder of the wall. [1]
 p) A steel cylinder of a car engine has an aluminium piston. The internal diameter of the cylinder of the wall. [1]
 q) A steel cylinder of a car engine has an aluminium piston. The internal diameter of the cylinder of the wall. [1]
 r) A steel cylinder of a car engine has an aluminium piston. The internal diameter of the cylinder of the wall. [1]
 s) A steel cylinder of a car engine has an aluminium piston. The internal diameter of the cylinder of the wall. [1]
 t) A steel cylinder of a car engine has an aluminium piston. The internal diameter of the cylinder of the wall. [1]
 u) A steel cylinder of a car engine has an aluminium piston. The internal diameter of the cylinder of the wall. [1]
 v) A steel cylinder of a car engine has an aluminium piston. The internal diameter of the cylinder of the wall. [1]
 w) A steel cylinder of a car engine has an aluminium piston. The internal diameter of the cylinder of the wall. [1]
 x) A steel cylinder of a car engine has an aluminium piston. The internal diameter of the cylinder of the wall. [1]
 y) A steel cylinder of a car engine has an aluminium piston. The internal diameter of the cylinder of the wall. [1]
 z) A steel cylinder of a car engine has an aluminium piston. The internal diameter of the cylinder of the wall. [1]
- 22) a) State and explain Coulomb's law in electrostatics. [3]
 b) Why coulomb's law is applicable to stationary charges only? [1]
 c) Define 1 coulomb charge on the basis of Coulomb's law. [1]
 d) Two charges 10^{-6}C and 10^{-7}C attract each other with a force of 400N. Calculate the distance between them if they are in i) air and ii) water (relative permittivity of water is 80) [1]

Best of Luck



KATHMANDU MODEL SECONDARY SCHOOL

First-Term Examination 2080

Class: XI 'Science'
Time: 3hrs.

SET - B

Subject: Physics '1011'

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

F.M.: 75

P.M.: 30

Attempt all the questions:

GROUP 'A' [11×1=11]

Multiple choice questions

- If pressure P, velocity V and time T are taken as fundamental physical quantities, the dimensional formula of force is
a) $[P^{-1}VT^2]$ b) $[PV^2T^2]$ c) $[P^{-1}V^2T^{-2}]$ d) $[PV^2T^2]$
- If the magnitude of difference of two-unit vectors is $\sqrt{3}$, then the magnitude of sum of the two vectors is
a) 1 b) 2 c) 3 d) $\sqrt{3}$
- Which of the following cannot be resultant of the vectors of the magnitude 8 and 4?
a) 2 b) 4 c) 6 d) 12
- A force of 20 N is applied at an angle of 60° to the vertical. What is the horizontal component of this force?
a) 20N b) 10N c) 17.32N d) 15N
- The maximum and minimum magnitude of the resultant of two given vectors is 17 units and 7 units respectively. If these two vectors are at right angles to each other, the magnitude of resultant is
a) 14 b) 16 c) 18 d) 18
- Y-component of velocity is 20 and x-component is 10. The direction of motion of the body with the horizontal at this instant is
a) $\tan^{-1} \frac{1}{2}$ b) 45° c) $\tan^{-1} 2$ d) 0°
- An ant moves towards the plane mirror with speed of 2m/s and the mirror is moved towards the ant with the same speed. What is the velocity of ant with respect to its image?
a) 4 m/s b) 8 m/s c) 16 m/s d) 16 m/s
- The focal length of concave mirror is f and the distance from the object to the pole is x. The magnification is
a) $\frac{f}{f-x}$ b) $\frac{x}{f}$ c) $\frac{f}{x}$ d) $\frac{f}{f+x}$
- A point object is placed on the axis of the concave mirror at a distance of 45 cm from the focal point of the mirror. Its image is formed at the point of object then focal length of the mirror is
a) 90 cm b) 135 cm c) 45 cm d) 15 cm
- Two equally charged, identical metal spheres A and B repel each other with force F. The spheres are fixed with a distance r between them. Third identical but uncharged sphere is brought in contact with A and then placed at midpoint of line joining A and B. The magnitude of net electrostatic force on third sphere is
a) 2F b) 3F/4 c) 3F/4 d) F/2
- A faulty thermometer with lower fixed point and upper fixed point 5°C and 95°C respectively. What is the correct temperature when the reading gives 35?
a) 60°C b) 40°C c) 33.3°C d) 20.5°C

GROUP 'B' [8×5=40]

Short answer questions:

- What are the limitations of the physical relation $\rho = \frac{3g}{4\pi r g}$ where symbols carry their usual meanings?
- Check the correctness of the system based on 20 cm, 1kg and $\frac{1}{2}$ minute as fundamental units.
- Find the value of 20 J on the system

(1)
(2)
(2)

GROUP 'A' MCQ [11x1=11]

1. Find the correct option of each question in your answer sheet.
 a) value of q required to b) Suppose harmonic means between two numbers 1 and 2 is 9×10^{22} kg
 B) $c = -1$ C) $c = m$ D) $cl=1$
 B) $\frac{2}{3}$ C) $\frac{3}{4}$ D) $\frac{4}{3}$
 2. relation between arithmetic mean (AM), geometric mean (GM) and harmonic mean (HM) between two unequal positive numbers is
 A) $GM > AM > HM$ B) $GM > AM > HM$ C) $AM > HM > GM$ D) $AM > HM > GM$
 3. Find the value of $\sqrt{-5} \times \sqrt{-6}$?
 A) $-\sqrt{30}$ B) $\sqrt{5} \times \sqrt{6}$ C) $i\sqrt{30}$ D) $i\sqrt{30}$
 4. 16 and 20 units or is the value of i^{281} ?
 A) -1 B) -1 C) i D) $-i$
 5. Inverse of the compound statement $p \Rightarrow q$ is
 A) $p \Rightarrow \sim q$ B) $\sim p \Rightarrow \sim q$ C) $\sim q \Rightarrow \sim p$ D) $p \Rightarrow \sim q$
 6. $(A \cap B)$ then
 A) $x \in A$ and $x \notin B$ B) $x \notin A$ or $x \notin B$ C) $x \in A$ and $x \in B$ D) $x \in A$ and $x \in B$
 7. Symmetric differences of two sets A and B is
 A) $(A \cup B) \cap (A \cap B)$ B) $(A \cup B) \cap (A \cap B)$ C) $(A - B) \cap (B - A)$ D) $(A - B) \cap (B - A)$
 8. Temperature of surrounding $(B) \cup (B - A)$
 A) $\frac{5x^2+2x-3}{4x^3+6x+5}$ is equal to
 B) 0 C) $\frac{5}{4}$ D) $\frac{4}{5}$
 9. $\frac{\tan^2 \alpha x}{\tan^2 \beta x}$ is equal to
 A) $\frac{\beta}{\alpha}$ B) $\frac{\alpha^2}{\beta^2}$ C) $\frac{\alpha^2}{\beta^2}$ D) $\frac{\beta^2}{\alpha^2}$
 10. $\frac{\tan x^0}{\sin x}$ is equal to
 A) π B) $\frac{\pi}{180}$ C) $\frac{\pi}{180}$ D) $\frac{180}{\pi}$

GROUP 'A' MCQ [11x1=11]

1. Re-write the correct option of each question in your answer sheet.
 1. If one root of $mx^2 - 3x + 2m - 5 = 0$ is zero then value of m is
 A) $m = \frac{5}{2}$ B) $m = -\frac{5}{2}$ C) $m = \frac{2}{5}$ D) $m = 1$
 2. The geometric mean between -8, -2 is
 A) 4 B) -4 C) 3 D) -2
 3. If three positive numbers a, b, c are in A.P then
 A) $b^2 > ac$ B) $b^2 = ac$ C) $b^2 < ac$ D) All of these
 4. What is the value of $\sqrt{-1} \sqrt{-4}$?
 A) 2i B) $\sqrt{(-4)(-1)}$ C) -2i D) -2
 5. What is the value of i^{383} ?
 A) i B) -i C) 1 D) -1
 6. The inverse of compound statements $q \Rightarrow p$ is
 A) $q \Rightarrow p$ B) $\sim p \Rightarrow \sim q$ C) $\sim q \Rightarrow \sim p$ D) $p \Rightarrow \sim q$
 7. If $x \in (A \cap B)$ then
 A) $x \in A$ or $x \in B$ B) $x \in A$ and $x \in B$ C) $x \notin A$ and $x \notin B$ D) $x \in A$ and $x \in B$
 8. The symmetric difference between two sets A and B is
 A) $A - B$ B) $(A \cup B) \cap (A \cap B)$ C) $(A \cup B) \cap (A \cap B)$ D) $(A - B) \cap (B - A)$
 9. What is the value of $\lim_{x \rightarrow \infty} \frac{x^3+3x+5}{5x^2+6x+8}$?
 A) $\frac{\infty}{\infty}$ B) ∞ C) $\frac{1}{5}$ D) 5
 10. What is the value of $\lim_{x \rightarrow 0} \frac{\sin x^0}{\tan x}$?
 A) $\frac{\pi}{180}$ B) $\frac{180}{\pi}$ C) 1 D) -1
 11. If $\lim_{x \rightarrow 0} \frac{e^{px}-1}{x} = 3$ then the value of p is?
 A) 3 B) $\frac{1}{3}$ C) 0 D) 1

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions:

GROUP 'B' [8x5=40]

12. a. Derive the dimensions of mass considering force, velocity and time as basic units. [2]
- b. A sphere of radius r moving through a fluid of density ρ with high velocity v experiences a retarding force of $F = k \rho^x v^y$, where k is dimensionless constant. Find the value of x, y and z so that establish the relation for retarding force. [2]
13. a. State parallelogram law of vector addition and derive the relation for magnitude and direction of resultant of two vectors using this law. [3]
- b. Two equal vectors have a resultant equal to $3/2$ times the value of either vector. Find the angle between them. [2]
14. Differentiate between heat and temperature. [1]
15. Write the cause and direction of heat flow. [1]
16. In an arbitrary scale of temperature called P scale, water freezes at $40^\circ P$ and boils at $290^\circ P$. Find the boiling point of a liquid in this P scale if it boils at $62^\circ C$. [1]
17. Coefficient of linear expansion depends on initial length and change in temperature of a substance? Explain. [3]
18. The marking of an aluminium ruler and brass ruler are perfectly aligned at how far apart will the 20 cm marks be on two rulers at $100^\circ C$ if precise alignment of left hand ends of the rulers is maintained. [2]
19. Aluminium and brass are $2.4 \times 10^{-5} C^{-1}$ and $2 \times 10^{-5} C^{-1}$ respectively. Use a mirror formula stating necessary sign convention. [3]
20. A concave mirror for face viewing has focal length of 0.4 m. Find the distance at which you hold the mirror from your face in order to see your upright with a magnification of 5. [2]
21. Explain Coulomb's law in electrostatics. [3]
22. Coulomb's law follows Newton's third law? Explain. [2]

18. a. Why pointed parts are avoided in electrical appliances? Explain.

- b. Two point charges of $5\mu C$ and $-80\mu C$ are separated by 24 cm in free space. Find the point along the line joining them at which net electrostatic force on any charge placed at that point is zero.

19. a. Define one coulomb charge.

- b. Why coulomb's law is applicable to the stationary charges only?

- c. Suppose a charge $+q$ on Earth's surface and another $+q$ charge is placed on the surface of the Moon. (a) Calculate the value of q required to balance the gravitational attraction between Earth and Moon (b) Suppose the distance between the Moon and Earth is halved, would the charge change? (Take $m_E = 5.9 \times 10^{24} \text{ kg}$, $m_M = 7.9 \times 10^{22} \text{ kg}$)

GROUP C (8x3=24)

20. a. What is meant by resolution of a vector? Find the rectangular components of a vector.
- b. Explain geometrical interpretation of the scalar product of vectors.
- c. There are three vectors of magnitude 8, 16 and 20 units oriented arbitrarily in the same plane. What are the minimum and maximum values of their resultant?
- d. Find angle between the vectors; $\vec{A} = 4\hat{i} + 3\hat{j}$, $\vec{B} = 5\hat{i} - 2\hat{j}$
21. a. Define linear expansivity. Describe Pullinger's method to find the linear expansivity of a solid.
- b. The coefficient of linear expansion of copper is $16 \times 10^{-6} C^{-1}$ calculate increase in length of a 50cm long copper wire when it is heated through $20^\circ C$.
- c. A second pendulum made of brass keeps correct time at $10^\circ C$. How much seconds it will lose or gain per day when the temperature of surroundings rises to $35^\circ C$. (α for brass $= 0.000018/C^\circ$)
22. a. Define radius of curvature and focal length of a curved mirror and derive relation between them.
- b. Explain the properties of image of a real object formed by a convex mirror with necessary ray diagram.
- c. An object 2 cm high is placed at a distance of 16 cm from a concave mirror which produces an image 3 cm high. What is the focal length of the mirror. Find the position of the image.

"Best of Luck"

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions:

GROUP 'A' MCQ [11x1=11]

1. The correct option of each question in your answer sheet is the value of $\sqrt{-5} \times \sqrt{-6}$?
A) $\sqrt{30}$ B) $-\sqrt{30}$ C) $\sqrt{5} \times \sqrt{6}$ D) $-\sqrt{5} \times \sqrt{6}$
2. Harmonic means between two numbers 1 and 2 is
A) $\frac{1}{3}$ B) $\frac{2}{3}$ C) $\frac{3}{4}$ D) $\frac{4}{3}$
3. Relation between arithmetic mean (AM), geometric mean (GM) and harmonic mean (HM) between two unequal positive numbers a and b is
A) $AM > GM > HM$ B) $GM > AM > HM$ C) $AM > HM > GM$ D) $GM > HM > AM$
4. Inverse of the compound statement $p \Rightarrow q$ is
A) $p \Rightarrow q$ B) $\sim p \Rightarrow \sim q$ C) $q \Rightarrow \sim p$ D) $\sim q \Rightarrow \sim p$
5. If $(A \cap B)$ then $(A \cup B)$ is
A) $A \cap B$ B) $A \cup B$ C) $A \cap B$ D) $A \cup B$
6. Symmetric differences of two sets A and B is
A) $A \Delta B$ B) $A \cap B$ C) $A \cup B$ D) $A \cap B$
7. $5x^2 + 2x - 3$ is equal to
A) $4x^2 + 6x + 5$ B) 0 C) $\frac{5}{4}$ D) $\frac{5}{4}$
8. $\frac{\tan^2 \alpha x}{\tan^2 \beta x}$ is equal to
A) $\frac{\alpha^2}{\beta^2}$ B) $\frac{\beta}{\alpha}$ C) $\frac{\alpha^2}{\beta^2}$ D) $\frac{\alpha^2}{\beta^2}$
9. $\frac{\tan x^0}{\sin x}$ is equal to
A) π B) π C) $\frac{\pi}{180}$ D) $\frac{\pi}{180}$

KATHMANDU MODEL SECONDARY SCHOOL
FIRST TERM EXAMINATION 2081

Class: XI 'Science'

Time: 25min.

Subject: Physics [1011]

Tick the best option among the four options. Apply tick on two options will be automatically denied and mark will not be counted.

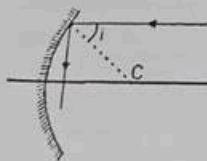
Choose the best alternative.

GROUP 'A' [11×1=11]

- The dimension $[ML^{-1}T^{-2}]$ corresponds to
(A) Work done by force (B) Pressure
(C) Linear momentum (D) Coefficient of linear expansion
- Parallelogram law of vectors is applicable to the addition of
(A) Any two vectors (B) Two scalars
(C) A vector and a scalar (D) Two vectors representing same physical quantity.
- Given: $\vec{A} = 2\hat{i} - 3\hat{j}$ and $\vec{B} = 5\hat{i} - 6\hat{j}$. The Magnitude of $(\vec{A} + \vec{B})$ is
(A) $\sqrt{120}$ units (B) $\sqrt{130}$ units
(C) $\sqrt{58}$ units (D) $\sqrt{65}$ units
- Vectors $\vec{A} = \hat{i} + \hat{j} - 2\hat{k}$ and $\vec{B} = 4\hat{i} + 4\hat{j} - 8\hat{k}$ are
(A) Parallel (B) Antiparallel
(C) Perpendicular (D) at acute angle with each other
Unit vector along $\hat{i} + \hat{j}$ is
(A) $\frac{\hat{i} + \hat{j}}{\sqrt{2}}$ (B) $\frac{3\hat{i} + 3\hat{j}}{\sqrt{2}}$
(C) $\hat{i} + \hat{j}$ (D) $\hat{k}$

If reflected ray intersects the principal axis perpendicularly then angle of incidence is

- (B) 30°
(D) 60°



- Convert 98.4°F into degree Celsius.
(A) 39°C (B) 36.9°C (C) 98.4°C (D) 374.4°C
- Two spheres one solid and another hollow both made of same material of same radius. If same heat is given to both spheres then expansion will be
(A) solid (B) hollow
(C) same (D) Depends on heating method
- A point charge $+Q$ is placed at the centroid of an equilateral triangle. A second charge $+Q$ is placed at a corner of the triangle, the magnitude of the electrostatic force on the central charge is 8 N. The magnitude of the force on the central charge when a third charge $+Q$ is placed at another corner of the triangle is:
(A) zero (B) 4 N (C) $4\sqrt{2}$ N (D) 8 N
- Two ebonite rods A and B of radii 1.5 cm and 2 cm respectively are rubbed together. The result will be
(A) Both remain neutral
(B) A will be charged positively
(C) A will be charged negatively
(D) Both will be charged equally similar charge.
- Two point charges placed at certain distance d in a medium of dielectric constant K exert a force F on each other. Then the distance at which these charges will experience the same force in air is
(a) dK (b) $d\sqrt{K}$ (c) $\frac{d}{\sqrt{K}}$ (d) $\frac{d}{K}$

KATHMANDU MODEL SECONDARY SCHOOL
FIRST TERM EXAMINATION 2081

SET B

Subject: Physics [1011]

Tick the best option among the four options. Apply tick on two options will be automatically denied and mark will not be counted.

Choose the best alternative.

GROUP 'A' [11×1=11]

- Find the dimension of K in the equation $W = \frac{1}{2} Kx^2$ and x is the extension produced.
(A) $[ML^0T^{-1}]$ (B) $[ML^0T^{-2}]$ (C) $[ML^2T^{-2}]$ (D) $[ML^2T^{-1}]$
- Vector along $3\hat{i} + 3\hat{j}$ is
(A) $\frac{3\hat{i} + 3\hat{j}}{\sqrt{2}}$ (B) $\hat{i} + \hat{j}$ (C) $\frac{3\hat{i} + 3\hat{j}}{2}$ (D) $\frac{3\hat{i} + 3\hat{j}}{3}$
- Vectors $\vec{A} = \hat{i} + \hat{j} - 2\hat{k}$ and $\vec{B} = 3\hat{i} + 3\hat{j} - 6\hat{k}$ are
(A) Parallel (B) Antiparallel
(C) Perpendicular (D) at acute angle
- Parallelogram law of vectors is applicable to the addition of
(A) Any two vectors (B) Two scalars
(C) A vector and a scalar (D) Two vectors representing same physical quantity
- Maximum magnitude of $(\vec{A} + \vec{B})$ will be the maximum magnitude of $(\vec{A} + \vec{B})$
(A) $A + B$ (B) $\sqrt{A^2 + B^2}$ (C) $A^2 + B^2$ (D) $\sqrt{A^2 + B^2}$
- Selected ray intersects the principal axis at a point. The distance from the optical center to this point is
(A) 30° (B) 60° (C) 45° (D) 90°
- Convert 69.8°F into degree Celsius.
(A) 30°C (B) 54°C (C) 45°C (D) 90°C

KATHMANDU MODEL SECONDARY SCHOOL
FIRST TERM EXAMINATION 2081

SET B

Subject: Physics [1011]

Best option among the four options. Apply tick on two options will be fully denied and mark will not be counted.

Best alternative.

GROUP 'A' [11×1=11]

(C) 98.4°C (D) 374.4°C
both made of same material
spheres then expansion

(B) hollow
(D) Depends on heating method
of an equilateral triangle.
the triangle, the magnitude
8 N. The magnitude of the
+ Q is placed at another

ector along $3\hat{i} + 3\hat{j}$ is

(D) 8 N

m respectively are rubbys $\vec{A} = \hat{i} + \hat{j} - 2\hat{k}$ and $\vec{B} = 3\hat{i} + 3\hat{j} - 6\hat{k}$ are

parallel

perpendicular

the law of vectors is applicable to the addition of

two vectors

no scalars

a medium of dielectric
distance at which the

vector and a scalar

no vectors representing same physical quantity.

will be the maximum magnitude of $(\vec{A} - \vec{B})$?

(d) $\frac{d}{K}$

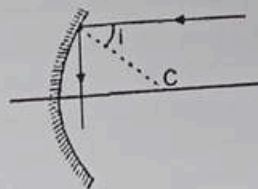
$\sqrt{A^2 + B^2}$

ected ray intersects the principal axis perpendicularly then angle of

ion is s

(B) 30°

(D) 60°



rt 69.8° F into degree Celsius.

°C

(B) 54° C

(C) 45° C

(D) 40° C

8. Two spheres one solid and another hollow both made of same material and of same radius. If both spheres are heated to the same temperature, then expansion will be more in

(A) solid

(B) hollow

(C) same

(D) Depends on heating method

9. A point charge + Q is placed at a corner of an equilateral triangle. When a second charge + Q is placed at another corner of the triangle, the magnitude of the electrostatic force on the first charge is 6 N. The magnitude of the net force on the first charge when a third charge + Q is placed at the remaining corner of the triangle is:

(A) zero

(B) 6 N

(C) $6\sqrt{3}$ N

(D) 8 N

10. Two identical metal spheres A and B are supported on insulating stands and placed in contact. What kind of charges will A and B develop when negatively charged ebonite rod is brought near A?

(a) Both will have positive charges

(b) Both will have negative charges

(c) A will have a positive charge and B will have negative charge

(d) B will have a positive charge and A will have negative charge

11. Two point charges placed at certain distance r in air exert a force F on each other. Then the distance at which these charges will experience the same force in a medium of dielectric constant K is

(a) r K

(b) $r\sqrt{K}$

(c) $\frac{r}{\sqrt{K}}$

(d) $\frac{r}{K}$

"Best of Luck"

KATHMANDU MODEL SECONDARY SCHOOL
FIRST-TERM EXAMINATION 2081

XI 'Science'
 3hrs.

Subject: Physics '1011'

F.M.: 75

P.M.: 30

Answers are required to give their answers in their own words as far as possible. The figures in the margin indicate full marks.

Attempt all the questions:

GROUP 'B' [8×5=40]

1. Derive the dimensions of mass considering force, velocity and time as basic units. [2]
2. A sphere of radius r moving through a fluid of density ρ with high velocity v experiences a retarding force of $F = k a^x \rho^y v^z$, where k is dimensionless constant. Find the value of x, y and z so that establish the relation for retarding force. [3]
3. State parallelogram law of vector addition and derive the relation for magnitude and direction of resultant of two vectors using this law. [3]
4. Two equal vectors have a resultant equal to $3/2$ times the value of either vector. Find the angle between them. [2]
5. Differentiate between heat and temperature. [1]
6. State the cause and direction of heat flow. [1]
7. On an arbitrary scale of temperature called P scale, water freezes at $40^\circ P$ and boils at $290^\circ P$. Find the boiling point of a liquid in this P scale if it boils at $120^\circ C$. [3]
8. How does coefficient of linear expansion depends on initial length and change in temperature of a substance? Explain. [2]
9. The marking of an aluminium ruler and brass ruler are perfectly aligned at one end. How far apart will the 20 cm marks be on two rulers at $100^\circ C$ if precise alignment of left hand ends of the rulers is maintained. [3]
10. The coefficient of linear expansion of aluminium and brass are $2.4 \times 10^{-5} C^{-1}$ and $2 \times 10^{-5} C^{-1}$ respectively. [3]
11. State a mirror formula stating necessary sign convention. [2]
12. A concave mirror for face viewing has focal length of 0.4 m. Find the distance at which you hold the mirror from your face in order to see your face upright with a magnification of 5. [3]
13. State and explain Coulomb's law in electrostatics. [2]
14. Does Coulomb's law follows Newton's third law? Explain. [2]

18. a. Why pointed parts are avoided in electrical appliances? Explain. [2]
- b. Two point charges of $5\mu C$ and $-80\mu C$ are separated by 24 cm in free space. Find the point along the line joining them at which net electrostatic force on any charge placed at that point is zero. [3]
19. a. Define one coulomb charge. [1]
- b. Why coulomb's law is applicable to the stationary charges only? [1]
- c. Suppose a charge $+q$ on Earth's surface and another $+q$ charge is placed on the surface of the Moon. (a) Calculate the value of q required to balance the gravitational attraction between Earth and Moon (b) Suppose the distance between the Moon and Earth is halved, would the charge q change? (Take $m_E = 5.9 \times 10^{24} \text{ kg}$, $m_M = 7.9 \times 10^{22} \text{ kg}$) [1]

GROUP C (8×3=24)

20. a. What is meant by resolution of a vector? Find the rectangular components of a vector. [3]
- b. Explain geometrical interpretation of the scalar product of two vectors. [2]
- c. There are three vectors of magnitude 8, 16 and 20 units oriented arbitrarily in the same plane. What are the minimum and maximum values of their resultant? [2]
- d. Find angle between the vectors; $\vec{A} = 4\hat{i} + 3\hat{j}$, $\vec{B} = 5\hat{i} - 2\hat{j}$ [1]
21. a. Define linear expansivity. Describe Pullinger's method to find the linear expansivity of a solid. [3]
- b. The coefficient of linear expansion of copper is $16 \times 10^{-6} C^{-1}$ calculate the increase in length of a 50cm long copper wire when it is heated through $20^\circ C$. [2]
- c. A second pendulum made of brass keeps correct time at $10^\circ C$. How many seconds it will lose or gain per day when the temperature of surrounding rises to $35^\circ C$. (α for brass = $0.000018/C^\circ$) [3]
22. a. Define radius of curvature and focal length of a curved mirror and derive relation between them. [1+2]
- b. Explain the properties of image of a real object formed by a convex mirror with necessary ray diagram. [2]
- c. An object 2 cm high is placed at a distance of 16 cm from a concave mirror, which produces an image 3 cm high. What is the focal length of the mirror? Find the position of the image. [3]

"Best of Luck"

Knowledge through Quality Education
KATHMANDU MODEL SECONDARY SCHOOL
THIRD TERM EXAMINATION 2081

Class: XI 'Science'

Time: 25min.

SET-A

Subject: Physics [1011]

Tick the best option among the four options. Apply tick on two options will be automatically denied and mark will not be counted.

Rewrite the best options of each question in your answer sheet.

GROUP 'A' [11×1=11]

1. Which of the following will reduce random errors in an experiment?
 A) Using an instrument having a greater precision
 B) Checking the calibration of the instrument used
 C) Checking for zero error on the instrument used
 D) Repeating readings
2. The resultant of two vectors $\vec{P}$ and $\vec{Q}$ is perpendicular to $\vec{P}$ and its magnitude is equal to the magnitude of the vector $\vec{Q}$. The angle between $\vec{Q}$ and $\vec{R}$ is
 A) 35° B) 45° C) 60° D) None of these
3. An object is dropped from the top of a tower of 120 m and at the same time another object is thrown vertically upwards with a velocity of 30 m/s from the foot of the tower. Find the time when balls will collide?
 A) 1 sec B) 2 sec C) 3 sec D) 4 sec
4. On heating a liquid having coefficient of volume expansion k in a container having coefficient of linear expansion $2k$, the level of the liquid in the container would
 A) rise B) fall C) remains almost stationary
 D) cannot be predicted
5. If a rod of length 2 m expands by 0.002 m when heated from 20°C to 100°C , what is the coefficient of linear expansion (α)?
 A) $1.25 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$ B) $2.5 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$
 C) $5.0 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$ D) $1.0 \times 10^{-4} \text{ }^\circ\text{C}^{-1}$
6. 600 J of heat raises the temperature of 20 g water. The temperature increase is:
 A) 7.14°C B) 14.28°C C) 3.57°C D) 28.57°C
7. A convex mirror always forms an image that is
 A) Real and inverted B) Virtual and diminished
 C) Virtual and same size as the object D) Real and enlarged

8. A concave mirror has a focal length of 10 cm. Where should an object be placed to get an image of the same size?
 A) 5 cm B) 10 cm C) 20 cm D) 30 cm
9. A diver underwater looks up at the surface at an angle of 40° to the normal. At what angle does an observer in the air see the ray emerging from the water?
 A) 49.8° B) 52.1° C) 55.4° D) 57.6°
10. The distance between two charges is increased by 25% then the percentage change in force is
 A) 36 % B) 56% C) 65% D) 50%
11. When separation between two charges is increased the electric potential energy of the system
 A) Increase B) Decrease
 C) may increase or decrease D) Remains same



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SET - B

Subject: Physics [1011]

Choose the best option among the four options. Apply tick on two options will be optically denied and mark will not be counted.
Choose the best options of each question in your answer sheet.

GROUP 'A' [11×1=11]

Which of the following is a valid statement?

- A measurement that is precise is always accurate.
- A measurement that is not precise can be accurate.
- A measurement that is not precise will always be inaccurate.
- Repeated measurements will always increase accuracy and precision.

The resultant of two vectors $\vec{P}$ and $\vec{Q}$ is perpendicular to $\vec{P}$ and its magnitude is equal to half of the magnitude of the vector $\vec{Q}$. The angle between $\vec{P}$ and $\vec{Q}$ is

- A) 120°
- B) 150°
- C) 135°
- D) None of these

An object is thrown vertically downward with the velocity 20 m/s from the top of a tower of 100 m and at the same time another object is thrown vertically upwards with a velocity of 30 m/s from the foot of the tower. Find the time when balls will collide?

- A) 1 sec
- B) 2 sec
- C) 3 sec
- D) 4 sec

On heating a liquid having coefficient of volume expansion k in a container having coefficient of linear expansion $k/4$, the level of the liquid in the container would

- A) rise
- B) fall
- C) remains almost stationary
- D) cannot be predicted

A rod of length 1.5 m expands by 0.0018 m when heated from 30°C to 100°C. What is the coefficient of linear expansion (α)?

- A) $8 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$
- B) $10 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$
- C) $12 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$
- D) $15 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$

100 g of water at 50°C is mixed with 200 g at 20°C. The final temperature

- A) 38°C
- B) 42°C
- C) 35°C
- D) 28°C

A concave mirror forms a real image of an object. The image is always inverted

- A) Erect
- B) Virtual
- C) Enlarged
- D) Diminished

8. A concave mirror has a focal length of 15 cm. Where should an object be placed to get an image of the same size?

- A) 7.5 cm
- B) 15 cm
- C) 30 cm
- D) 45 cm

9. A fish appears to be at a depth of 2 m when viewed from above the water. If the refractive index of water is 1.33, what is the actual depth of the fish?

- A) 1.50 m
- B) 2.66 m
- C) 2.75 m
- D) 3.00 m

10. If the distance between two charges is increased by 50%, the percentage change in the electrostatic force is:

- A) 36% decrease
- B) 56% decrease
- C) 65% decrease
- D) 50% decrease

11. When separation between two charges is decreased the electric potential energy of the system

- A) Increase
- B) Decrease
- C) may increase or decrease
- D) Remains same